

MYE Practical Exam Mark Scheme

Qn	Skills assessed	Marking scheme	Mark	Mark ref
1(a)(i)	PDO Layout P1	Tabulates temperature and volume of FA 2 added in the table. Table has correct headers and units.	1	1
	PDO Record P2	All the volumes are recorded to the nearest 0.05 cm ³ and temperature to 0.1 °C	1	2
(a)(ii)	PDO Layout P3	Axes correct way round + correct labels + units + scale. Sensible linear scale must be chosen so that the plotted points occupy at least half the graph grid in both x and y directions.	1	3
	PDO Layout P4	All points correctly plotted to within $\pm 1/2$ small square. Checks all points and put ticks if correct.	1	4
	PDO Mani P5	Graph line must be best fit-lines. Lines should be extrapolated until they cross. Do not allow mark if clearly anomalous points are included or if the lines do not cross each other	1	5
(a)(iii)	PDO Mani P6	Correct reading to within $\pm 1/2$ small square of T_{maximum} . Correct calculation of $\Delta T_{\text{maximum}}$. Correct reading of $V_{\text{equivalence}}$.	1	6
	MMO Quality M1 M2	Award MR 7 based on the difference, $\Delta T_{\text{maximum}}$, between student's and supervisor's $\Delta T_{\text{maximum}}$ value. Give 1 mark if this difference is ≤ 0.3 °C Award MR 8 based on the difference, $V_{\text{equivalence}}$, between student's and supervisor's $V_{\text{equivalence}}$ value. Give 1 mark if this difference is ≤ 0.75 cm ³ Shift 1: Jessie 10.75 cm ³ and 5.1 °C, Kelly 10.40 cm ³ and 4.88 °C Shift 2: Joseph 10.33 cm ³ and 5.3 °C, Russell 10.00 cm ³ and 5.7 °C (Physics) Shift 3: Jessie 12.40 cm ³ and 4.85 °C (Biology), Russell 12.00 cm ³ and 6.7 °C (Physics)	1 1	7 8

(b)(i)	ACE Interpret A1	Calculates percentage error for 40 cm ³ of FA 1 using measuring cylinder and burette % error (measuring cylinder) = $(0.5 / 40) \times 100 \%$ % error (burette) = $[(2 \times 0.05) / 40] \times 100 \%$	1	9
(b)(ii)	ACE Interpret A2	States that the apparatus is burette and explains that NaHCO ₃ is the limiting reagent thus accuracy is essential.	1	10
(b)(iii)	PDO Layout P7	Records T _{FA1} , T _{FA3} and T _{mixture} to 0.1 °C.	1	11
(c)(i)	ACE Interpret A3	$[\text{NaOH}] = (25.0 \times 10^{-3} / V_{\text{equivalence}} \times 10^3) \times 1.00$	1	12
(c)(ii)	ACE Interpret A4 A5	$Q = (V_{\text{equivalence}} + 25.0)(1.00) c \Delta T_{\text{maximum}}$ $n(\text{NaHCO}_3) = 1.00 \times 25.0/1000 = 0.025 \text{ mol}$ $\Delta H_{\text{reaction1}} = -q / n(\text{NaHCO}_3)$	1	13
			1	14
(d)	ACE Interpret A6 A7	$\Delta T_{\text{maximum2}} = T_{\text{mixture}} - T_{\text{average}} $ $q = (V_{\text{FA 1}} + V_{\text{FA3}}) \times 4.18 \times \Delta T_{\text{maximum2}}$	1	15
		$\Delta H_{\text{reaction2}} = +q / 40.0 \times 10^{-3}$	1	16
	PDO Display P8	Shows working in all calculations in 1(a), 1(b), 1(c) and 1(d) All calculations must be relevant although they may not be complete or correct	1	17
	PDO Display P9	Shows appropriate significant figures (3 or 4) in all final answers in 1(a), 1(b), 1(c) and 1(d)	1	18
	PDO Display P10	Shows appropriate units in all final answers in 1(a), 1(b), 1(c) and 1(d)	1	19

(e)	ACE Interpret A8 A9 A10	$\begin{array}{c} \Delta H_{r3} \\ \text{CO}_2 + 2\text{H}_2\text{O} + \text{CO}_3^{2-} \rightarrow 2\text{HCO}_3^- + \text{H}_2\text{O} \\ \Delta H_{r2} \uparrow \qquad \qquad \qquad \uparrow \Delta H_n \\ \text{HCO}_3^- + \text{H}^+ + \text{CO}_3^{2-} + \text{H}_2\text{O} \leftarrow 2\text{HCO}_3^- + \text{H}^+ + \text{OH}^- \\ \Delta H_{r1} \end{array}$ Each part of cycle not balanced deduct 1 mark $\Delta H_{\text{reaction}3} = (-57.1) + \text{-(c)(ii)} + \text{-(d)}$	2	20 21
			1	22
2(a)	PDO Layout P11	Tabulates volume of FA 6 , water and time taken for blue colour to appear in the table. All the volumes are recorded to 1 d.p. and time to nearest 0.1 s.	1	21
	ACE Interpret A11	Adds volume of water to maintain a constant total volume of 71 cm ³ .	1	22
	ACE Interpret A12	Correctly calculates $V_{\text{FA6}} \times t$ and $V_{\text{FA6}}^2 \times t$.	1	23
(b)(i)	ACE Interpret A13	Explains that the same volume of $\text{S}_2\text{O}_3^{2-}$ is used in each experiment, thus same amount, thus amount of I_2 reacted is the same which gives a constant concentration of I_2 .	1	24
(b)(ii)	ACE Conclusion A15	Explains that $V_{\text{FA6}} \times t$ is constant for experiment 1 to 3, $n = 1$. Thus, order of reaction w.r.t bromate(V) is 1.	1	25
(c)	ACE Interpret A16	Correctly calculates $V_{\text{FA5}} \times t$ and $V_{\text{FA5}}^2 \times t$ and correctly deduces the order of reaction with respect to H^+ .	1	26
	ACE Conclusion A17	Explains that $V_{\text{FA5}}^2 \times t$ is constant for experiment 1 to 3, $n = 2$. Thus, order of reaction w.r.t H^+ is 2.	1	28
(d)	ACE Conclusion	States that is the statement is not valid and explains that to calculate half-life, concentration of sulfuric acid and bromate(V) has to be in large excess. However, the concentration of	1	29

	A19	bromate(V) is not in large excess (x10).		
(e)	ACE Interpret A20	States that excess I_2 left in the flask will react with the $S_2O_3^{2-}$ in the new experiment. As there is lesser $S_2O_3^{2-}$ for the new experiment, less I_2 is reacted, the time taken to produce the I_2 will be shorter in the second experiment and hence observe the blue colour faster. or State that the excess chemicals (H^+ , BrO_3^- , I^-) left in the flask will increase the concentrations of (H^+ , BrO_3^- , I^-) in the next reaction. This will cause the rate of reaction to increase and the time taken to produce the I_2 will be shorter and hence observe the blue colour faster.	1	30
(f)(i)	Plan PI 1 PI 2	Suggest appropriate volumes of H^+ , $KBrO_3$ and KI to make a mixture of total volume 250 cm^3 . Ensures that $[KI]$ is at least 10x lesser than $[H^+]$ and $[KBrO_3]$.	1 1	31 32
(ii)	ACE Interpret A21 A22	States that colour is blue and/or purple Explains that since the observed colour of iodine is yellow-orange, the complimentary colour is absorbed.	1 1	33 34
(iii)	PDO Layout P8 ACE Interpret A23	Axes correct way round + correct labels + units + scale + labeling of 50, 75 and 100% $[I_2]$. 1 st order product time graph with 2 constant consecutive half-lives.	1 1	35 36
	ACE Conclusion A24	States that the order of reaction with respect to $[I^-]$ will have constant half-lives.	1	37
3(a)(i)	MMO/ PDO Collecting	7 observation points – 2 with negative results Observations correctly recorded		

	M4 M5	6-7 = 3 4-5 = 2 2-3 = 1 0-1 = 0	3	38 39 40
(ii)	ACE Conclusion A25 A26	Argues for FA 8 is ethanal as it is the only one to form a silver mirror as it is oxidised by Tollens' reagent or the only one to form brick red precipitate as it is oxidised by Fehling's solution.	1	41
		Argues for FA 9 is propan-2-ol as it is oxidised to decolourise potassium manganate(VII) or oxidised to form a yellow precipitate by aqueous iodoine but not oxidised by Tollens' or Fehling's.	1	42
(b)(i)	ACE Interpret A27	Correctly identifies the structural feature as $\begin{array}{c} \text{OH} \\ \\ \text{R}-\text{C}-\text{CH}_3 \\ \\ \text{H} \end{array}$ and/or $\begin{array}{c} \text{O} \\ \\ \text{R}-\text{C}-\text{CH}_3 \end{array}$ where R = H or alkyl group. Labelling of R is optional.	1	43
(b)(ii)	ACE Interpret A28	Give the correct structural formula $\begin{array}{c} \text{O} \\ \\ \text{H}-\text{C}-\text{CH}_3 \end{array}$	1	44
(c)(i)	ACE Conclusion A29	Suggests Al^{3+} , Ba^{2+} , Mg^{2+} , Zn^{2+} (Ca^{2+} and Mn^{2+}) as cations that may be present.	1	45
(c)(ii)	Plan PI 3	Selects NaOH(aq) and NH_3 (aq) for identification of cation.	1	46
	MMO/ PDO Collecting M6 M7	Gives observation of NaOH(aq) with cation. <u>White precipitate</u> with NaOH(aq) dropwise for <u>Al^{3+}, Mg^{2+} and Zn^{2+}</u> . <u>White precipitate dissolves in excess NaOH(aq) for Al^{3+} and Zn^{2+}</u> . [1]	1	47
		Gives observation of NH_3 (aq) with cation. <u>White precipitate</u> with NH_3 (aq) dropwise for <u>Al^{3+} and Zn^{2+}</u> . <u>White precipitate dissolves in excess NH_3(aq) for Zn^{2+} but not Al^{3+}</u> [1]	1	48
	ACE	Concludes that FA 10 contains Zn^{2+} .	1	49

	Conclusion A30			
3(d)	Plan	Tests include (a) adding Br ₂ (aq), Br ₂ (l) or cold dilute alkaline KMnO ₄ (b) AgNO ₃ (aq) (c) hot acidified K ₂ Cr ₂ O ₇ (d) heat with NaOH(aq), cool, HNO ₃ (aq), followed by AgNO ₃ (aq)		
	PI 4	Selects reagents that would unambiguously identify 2 compounds with positive tests for each. [1]	3	50
	PI 5	or Selects reagents that would unambiguously identify 3 compounds with positive tests for each. [2]		51
	PI 6	or Selects reagents that would unambiguously identify 4 compounds with positive tests for each. [3]		52
	PI 7	Outlines a logical sequence/order of testing; doesn't test compounds already identified [1]	1	53
	PI 8	Some ambiguity in links etc, but worthy of credit [1]	2	54
	PI 9	or Clearly and unambiguously links the testing sequence with the compounds to be identified [2]		55